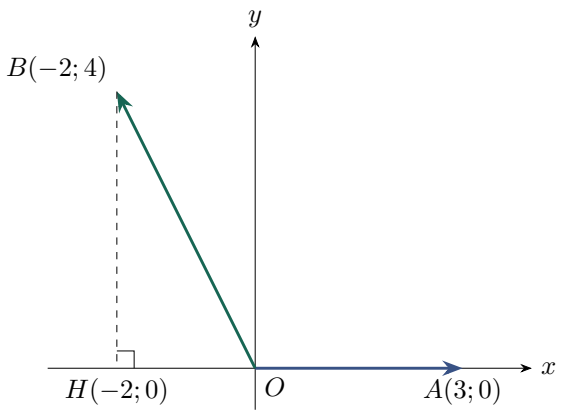
3.1 Projection and angle

Definition and results

For two nonzero vectors $\vec{u}, \vec{v}$, their scalar product is the real number $\vec{u} \cdot \vec{v} = \|\vec{u}\| \, \|\vec{v}\| \cos\theta$, where $\theta \in [0; \pi]$ is their unoriented angle. With a zero vector, the product is 0. If H is the projection of B onto (OA), $\overrightarrow{OA} \cdot \overrightarrow{OB}$ equals $OA \times OH$ when H is on the ray $[OA)$, and its opposite otherwise.

Status and justification. Definition; the equivalence with the projection follows from the cosine in right triangles, with a negative algebraic projection for an obtuse angle.

Worked example. For two vectors of norms 3 and 4 forming 60°, the product is $\dfrac{3 \times 4 \times 1}{2} = 6$. With 120°, it is -6. If $A(3;0)$ and $B(-2;4)$, the projection is $H(-2;0)$ and the product is -6.



The projection H lies on the negative side of the horizontal axis. Hence $\overrightarrow{OA} \cdot \overrightarrow{OB} = 3 \times (-2) = -6$.

Watch out

The result is a number, not a vector. A right angle corresponds to 0, not to 1.

Immediate application. Two vectors have norms 2 and 5 and form 90°, then 180°. Calculate the products.

3.2 Coordinates, norm and orthogonality

Definition and results

In an orthonormal basis, for $\vec{u} = (x; y)$ and $\vec{v} = (x\prime; y\prime)$, $\vec{u} \cdot \vec{v} = xx\prime + yy\prime$ and $\|\vec{u}\|^2 = x^2 + y^2$. The vectors are orthogonal if and only if their product is zero. For an orthonormal basis $(\vec{i}, \vec{j})$, $x = \vec{u} \cdot \vec{i}$ and $y = \vec{u} \cdot \vec{j}$.

Status and justification. Properties proved using bilinearity: expand $(x\vec{i} + y\vec{j}) \cdot (x\prime\vec{i} + y\prime\vec{j})$ and use $\vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = 1$, $\vec{i} \cdot \vec{j} = 0$.

Worked example. $\vec{u} = (2; 3)$, $\vec{v} = (3; -2)$: product $6 - 6 = 0$. The two vectors are orthogonal. Their norm is $\sqrt{13}$. The product of $(2; 3)$ with $(1; 0)$ is 2: we recover the x-coordinate.

Watch out

The usual formula assumes an orthonormal basis. Collinearity and orthogonality are different properties.

Immediate application. For which real number t are the vectors $(t; 2)$ and $(3; -6)$ orthogonal?

3.3 Vector calculus and identities

Definition and results

The scalar product is symmetric and bilinear: $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$ and $(\lambda\vec{u}) \cdot \vec{v} = \lambda(\vec{u} \cdot \vec{v})$. Hence $\|\vec{u} \pm \vec{v}\|^2 = \|\vec{u}\|^2 \pm 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2$. It follows that $\vec{u} \cdot \vec{v} = \dfrac{\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2}{2}$.

Status and justification. Bilinearity accepted from the geometric definition; symmetry comes from the cosine. The identities are proved by expanding $(\vec{u} \pm \vec{v}) \cdot (\vec{u} \pm \vec{v})$.

Worked example. If $\|\vec{u}\| = 3$, $\|\vec{v}\| = 4$ and $\|\vec{u} + \vec{v}\| = 5$, the product is $\dfrac{25 - 9 - 16}{2} = 0$. We deduce orthogonality without coordinates.

Watch out

In general, $\|\vec{u} + \vec{v}\| \neq \|\vec{u}\| + \|\vec{v}\|$. The identities concern the squares of the norms.

Immediate application. Given $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$.

3.4 Al-Kashi and choosing a method

Definition and results

In a triangle ABC, $BC^2 = AB^2 + AC^2 - 2AB\,AC\cos\widehat{BAC}$. This formula allows us to calculate the third side from two sides and the included angle, or to find the cosine of an angle from the three sides.

Status and justification. By Chasles, $\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB}$. Expanding its scalar square gives $BC^2 = AC^2 + AB^2 - 2\overrightarrow{AB}\cdot\overrightarrow{AC}$; then introduce the expression with the cosine.

Worked example. $AB = 5$, $AC = 3$ and $\widehat{BAC} = 60°$: $BC^2 = 25 + 9 - 15 = 19$, so $BC = \sqrt{19}$. With three sides, isolate the cosine before approximating the angle on a calculator, if one is allowed.

Watch out

The angle is included between the two sides of the product term. The minus sign is essential.

Immediate application. A triangle has $AB = AC = 4$ and $BC = 4\sqrt{2}$. Determine the angle A.

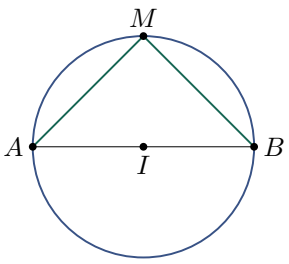
3.5 The circle with diameter AB

Definition and results

For two distinct points A and B with midpoint I, $\overrightarrow{MA}\cdot\overrightarrow{MB} = MI^2 - \dfrac{AB^2}{4}$. The set of points M such that $\overrightarrow{MA}\cdot\overrightarrow{MB} = 0$ is the circle with centre I and radius $\dfrac{AB}{2}$, including the points A and B.

Status and justification. Write $\overrightarrow{MA} = \overrightarrow{MI} + \overrightarrow{IA}$ and $\overrightarrow{MB} = \overrightarrow{MI} - \overrightarrow{IA}$. The cross terms cancel; what remains is $MI^2 - IA^2$. Since $IA = \dfrac{AB}{2}$, equality to zero is equivalent to the metric equation of the circle.

Worked example. $A(-2;0)$ and $B(2;0)$: I is the origin. The locus is $x^2 + y^2 = 4$. For $M(0;1)$, the product is $1 - 4 = -3$; for $M(0;2)$, it is 0.



Circle with diameter AB: the vectors $\overrightarrow{MA}$ and $\overrightarrow{MB}$ are orthogonal.

Watch out

The endpoints belong to the locus even though “right triangle” does not apply to a degenerate triangle.

Immediate application. For $A(0;0)$, $B(6;0)$, determine the points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$.